

Rohit Menon

Robotics Researcher | Perception, Reasoning and Manipulation

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[Website](#) | [Google Scholar \(242 citations; h-index: 9; i10-index: 9\)](#) | [GitHub](#) | [LinkedIn](#)

Profile

- Robotics researcher and PhD candidate in **deep learning, scene-graph reasoning, active perception and mobile manipulation**
- Former **Neura Robotics founding engineer and AI group leader**; independently built MAiRA's perception-manipulation architecture and developed its hard real-time control

Professional Experience

Research Associate

Feb. 2022–Present

Humanoid Robots Lab, University of Bonn, Germany

Research & Development

- Developed **SG-DOR: scene-graph reasoning with graph neural networks** to infer plant-part attachments and direction-conditioned occlusions; first author, IROS 2026
- Developed **O-STaR: open-vocabulary mobile object search using LLM priors, geometric reasoning and probabilistic belief updates** on dynamic scene graphs; submitted to ICRA 2027
- Co-developed **HortiBot's three-arm harvesting system**, integrating active perception, online plant mapping and manipulation; equal-contribution author, IROS 2024
- Developed **NBV-SC: shape-completion-guided view planning** for fruit mapping and reconstruction; validated with a UR5e in a commercial greenhouse; first author, IROS 2023
- Developed **EvidMTL and EvidKimera: evidential deep learning for depth, semantics and uncertainty quantification** in 3D semantic surface mapping; first author, IROS 2025
- Co-developed **Dawn-IK: optimization-based, collision-aware inverse kinematics** for heterogeneous multi-arm systems; equal-contribution author, Humanoids 2023

Hardware & Systems

- Designed and procured hardware for **HortiBot**, a Neobotix-based **omnidirectional mobile manipulator**, and the **PhenoRob Phase 2 quadruped and sensors**
- Annotated **tomato and pepper plant datasets**; generated **synthetic data** for robotic perception

Leadership & Outreach

- Supervising **three junior PhD students**; supervised **five MSc theses**, yielding **two full papers and one workshop paper**; two theses received awards in 2026
- Assisted in writing the **Agriloop proposal (EUR 400,000)** for **foundation model adaptation for human-in-the-loop robotic intervention in greenhouses**
- Contributed to the **PhenoRob Phase 2 HRL PhD proposal**; served on the **PhD interview committee**
- Member of the **RIG semantic perception cluster**; organizing the **PhenoRob Photosynthesis Study Group**; redesigned Humanoid Robotics teaching and delivered Cognitive Robotics lectures and exams; demonstrated robots on **WDR Bonn**, to the **NRW State Premier** and at **Wissenschaftsfest am Hofgarten**

Founding Engineer / Group Leader, AI for Manipulation

Nov. 2019–Jan. 2022

Neura Robotics GmbH, Metzingen, Germany

- **Independently built MAiRA's perception-manipulation architecture** as the AI team's first software engineer, including C++/ROS drivers and communication interfaces
- Rewrote Xenomai/Orocos RTT tasks for **hard real-time control and actuator communication** via SOEM EtherCAT
- Developed selective pick-and-place using semantic segmentation and depth fusion; integrated **voice- and gesture-based commands** into robot manipulation
- Built **human collision avoidance with RGB, depth and radar fusion**, combining multimodal sensing with real-time 3D human pose tracking
- Led **AI for Manipulation** after promotion in September 2021; held **hiring responsibility**; demonstrated robots at GITECH 2021, with coverage in *The National*

Earlier Experience

Researcher, Robot Manipulation and Control

July 2015–Oct. 2019

DFKI Robotics Innovation Centre, Bremen, Germany

- Solely responsible for **modelling, control and communication** of NeTTUN's tunnel-boring maintenance manipulator, including remote teleoperation
- Led **DFKI's iLEVATOR contribution**, managing budget, procurement and sponsor reporting; the mobile bin-picking system was **tested and deployed at the end user's factory**
- Independently developed **jerk-limited trajectory generation for high-DoF systems**; built an **unknown-object picking pipeline** combining deep-learning grasping with optimization-based motion planning
- Developed mobile-manipulator planning for **Hybr-iT** and **reactive grasping and upper-limb motion planning for the RH5 humanoid** in TransFit

MSc Thesis Student, Shared Autonomy

June 2014–Feb. 2015

DLR Institute of Robotics and Mechatronics, Oberpfaffenhofen, Germany

- Developed **adaptive shared autonomy** for assistive manipulation, combining **automatic grasp selection and virtual-fixture guidance** while preserving user control of the approach direction
- Validated the method in **full-dynamics simulation, a user study and on the real robot**; published at ICRA 2016

Engineering and Commissioning / Senior Executive

Sept. 2008–Sept. 2012

Siemens Ltd, Thane, India

- Developed **drive controls and PLC/HMI software**; commissioned machinery and tuned control loops.

Education

PhD in Computer Science

Feb. 2022–Mar. 2027 (Expected)

University of Bonn, Germany | Advisor: [Prof. Dr. Maren Bennewitz](#)

MSc in Automation and Robotics

Oct. 2012–Mar. 2015

TU Dortmund University, Germany | Grade: 1.2; top 5% of cohort

Bachelor's in Electrical Engineering

Aug. 2004–June 2008

SPCE, University of Mumbai, India | Grade: 76%; silver medal

Skills

Research: Deep learning; graph neural networks; shape completion; scene graphs; uncertainty quantification; 3D semantic mapping; active perception; open-vocabulary mobile manipulation

Programming and learning: C++, Python, MATLAB, PyTorch, PyTorch Geometric

Robotics and simulation: ROS, ROS 2, Isaac Sim, Gazebo, Blender

Control and libraries: Xenomai, Orocos RTT, SOEM EtherCAT, OpenCV, PCL, Eigen, NLopt

Languages: English C1 (TOEFL: 116/120), German B1, Hindi, Malayalam

Selected Awards

Best Poster Award, IROS 2025 Workshop on Perception and Planning for Mobile Manipulation in Changing Environments.

Scholarships: Deutschlandstipendium (2013), Ratan Tata scholarship (2006, 2007).

Industry award: Siemens Excellence in Commissioning trophy (2011).

Selected Publications

1. **Rohit Menon***, Niklas Mueller-Goldingen*, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz "**SG-DOR**: Learning Scene Graphs with Direction-Conditioned Occlusion Reasoning for Pepper Plants." *IROS*, 2026 (accepted)
2. **Rohit Menon**, Yasmin Schmiede, Gokul Krishna Chenchani, Hermann Blum, Maren Bennewitz "**O-STaR**: Effective Object Search through Spatio-Temporal Reasoning on Dynamic Scene Graphs." *GRC*, 2026 (Extended version under review, *ICRA*, 2027)
3. **Rohit Menon**, Nils Dengler, Sicong Pan, Gokul Krishna Chenchani, Maren Bennewitz "**EvidMTL**: Evidential Multi-Task Learning for Uncertainty-Aware Semantic Surface Mapping from Monocular RGB Images." *IROS*, 2025
4. Christian Lenz*, **Rohit Menon***, Michael Schreiber, Melvin Paul Jacob, Sven Behnke, Maren Bennewitz "**HortiBot**: An Adaptive Multi-Arm System for Robotic Horticulture of Sweet Peppers." *IROS*, 2024
5. Salih Marangoz*, **Rohit Menon***, Nils Dengler, Maren Bennewitz "**DawnIK**: Decentralized Collision-Aware Inverse Kinematics Solver for Heterogeneous Multi-Arm Systems." *Humanoids*, 2023
6. **Rohit Menon**, Tobias Zaenker, Nils Dengler, Maren Bennewitz "**NBV-SC**: Next Best View Planning based on Shape Completion for Fruit Mapping and Reconstruction." *IROS*, 2023
7. Jörn Vogel, Katharina Hertkorn, **Rohit U. Menon**, Máximo A. Roa. "**Flexible, Semi-Autonomous Grasping for Assistive Robotics**." *ICRA*, 2016

*Equal contribution.